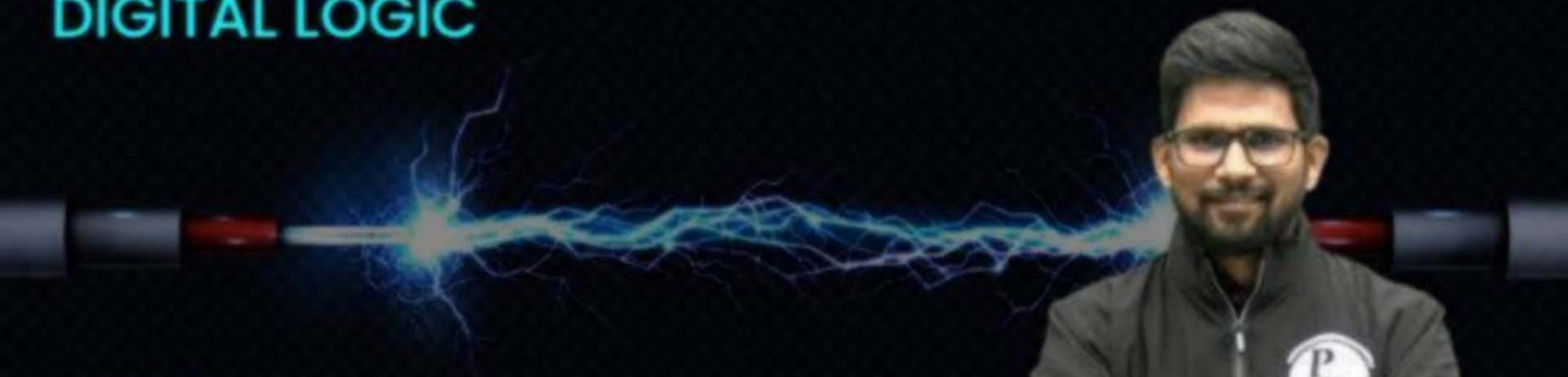
COMPUTER SCIENCE & IT

DIGITAL LOGIC



Lecture No. 08

BOOLEAN THEOREMS AND GATES



By- Chandan Gupta Sir



XOR & XNOR gate



- NAND gate

H.W.

Q.1 $A\bar{B} \oplus \bar{B}C \oplus (\bar{A} + B + \bar{C})$ is equal to Ans. A.

a. $B + \bar{A}\bar{C}$ b. $\bar{B} + AC$ c. $AB\bar{C}$ d. $A\bar{B}C$

Q.2 $A \oplus (\bar{A} + \bar{B})$ is equal to ⟶ Ans (c)

a. $A\bar{B}$ b. $\bar{A}B$ c. $\bar{A} + B$ d. $A + \bar{B}$

Q.3 $\bar{A} \oplus \bar{B} \oplus \bar{A}\bar{B} \oplus ABC$ is equal ⟶ Ans. (c)

a. $\bar{A} + \bar{B} + \bar{C}$ b. $A + B + C$ c. $\bar{A} + \bar{B} + C$ d. $(AB + BC)$

Q.4. $(A \oplus \bar{B}) \oplus (C \oplus \bar{B}) \oplus (B \oplus \bar{A}) \oplus (\bar{A} \oplus \bar{C})$ is equal to

a. $A \oplus B \oplus C$ b. $A \odot B \oplus C$ c. $(A \odot B)$ d. $(A \oplus B)$ ⟶ Ans. (c)

Q.5 $f(A, B, C, D)$

$= AC\bar{D} + \bar{B}\bar{D} + \bar{A}\bar{C} + ABC$

$= \Sigma(\ ?\ )$

Q.6 $f(A, B, C)$

$= A + \bar{B}\bar{C} + \bar{A}C$

$= \Sigma(\ ?\ )$

Q.5 ⟶ Ans. $\Sigma(0, 1, 2, 4, 5, 8, 10, 14, 15)$

Q.6 Ans: $\Sigma(0, 1, 3, 4, 5, 6, 7)$

Q.7 $f(A,B,C) = \underline{(B+\bar{C})}\,(\bar{A}+\bar{B})(A+\bar{B}+C) = \Pi(?) = \Sigma(?)$

Ans. $\Pi(1,2,5,6,7)$

$\Sigma(0,3,4)$

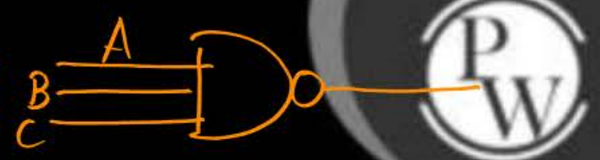
	B	$\bar{C}$	
0	0	$1 \rightarrow 1$	} POS
1	0	$1 \rightarrow 5$	} POS

[NAND GATE]

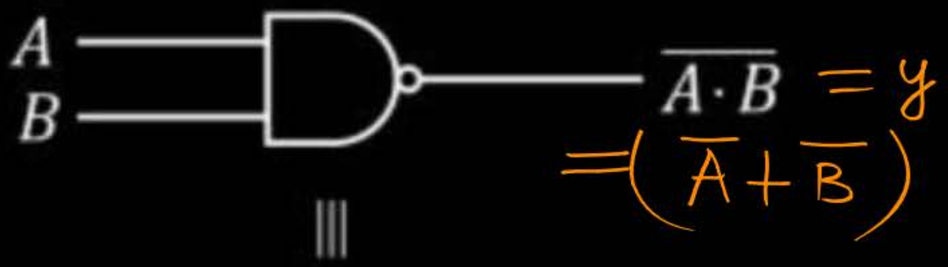
AND + NOT



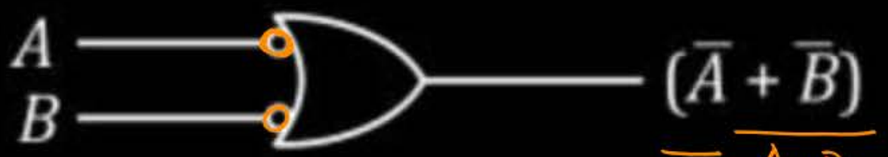
$\equiv$

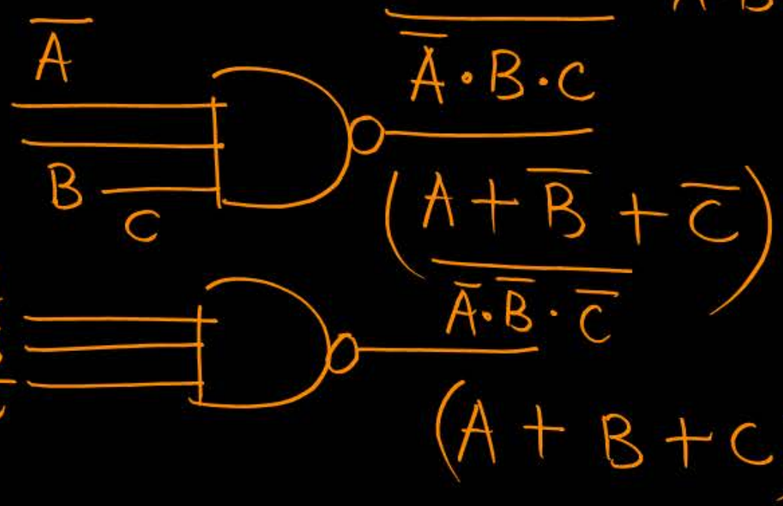


Representation :



$\equiv$





	2^1	2^0	
	A	B	$y = \overline{A \cdot B}$
0	0	0	1
1	0	1	1
2	1	0	1
3	1	1	0

$y(A, B) = \pi(3)$

$\equiv (0, 1, 2) = (\overline{A} + \overline{B})$

$= \overline{A}\,\overline{B} + \overline{A}B + A\overline{B}$

$= \overline{A} + (A \cdot \overline{B})$

$= (\overline{A} + A) \cdot (\overline{A} + \overline{B})$

$= (\overline{A} + \overline{B})$

$= \overline{A \cdot B}$

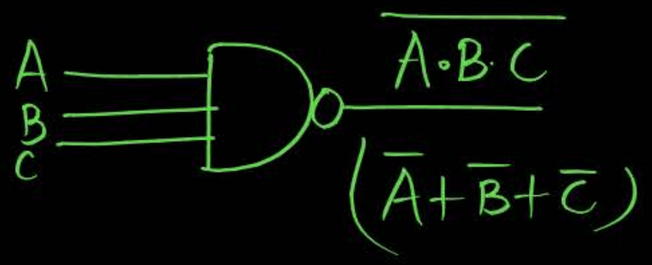
- Commutative Law :



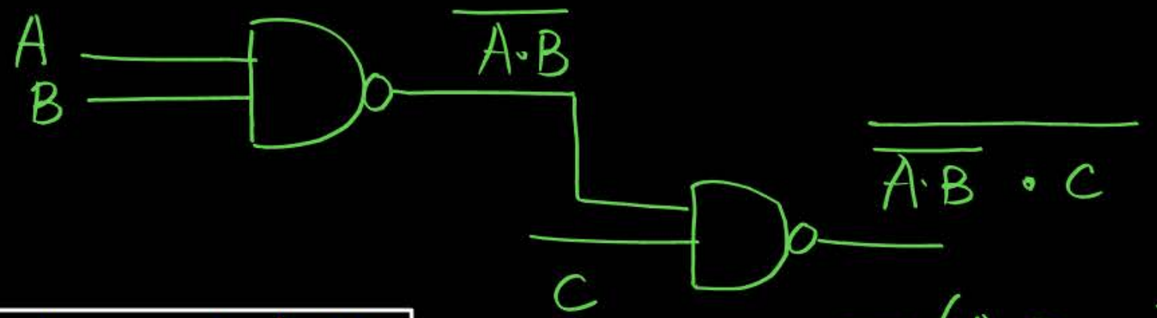
It holds commutative law.

Position of variables is irrelevant.

- Associative Law :



$\neq$



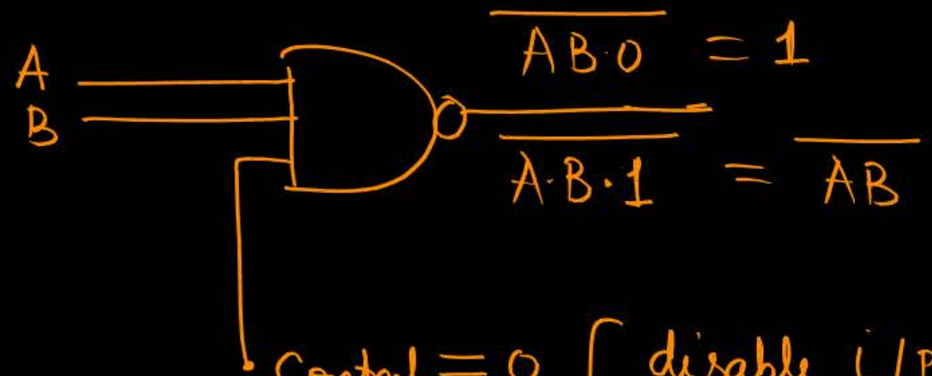
$= (A \cdot B + \bar{C})$

It does not hold associative law.

- IMP Points :

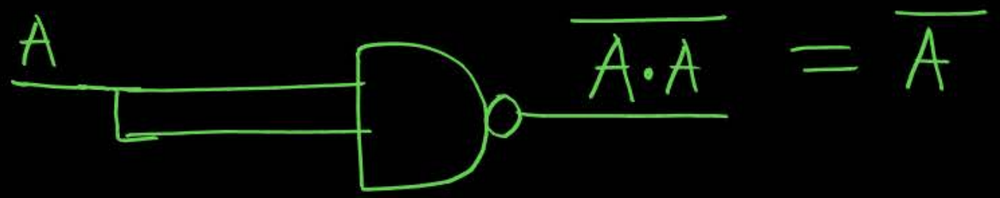
- If any one of i/p line is '0' then irrespective of other i/p lines o/p will definitely be '1'

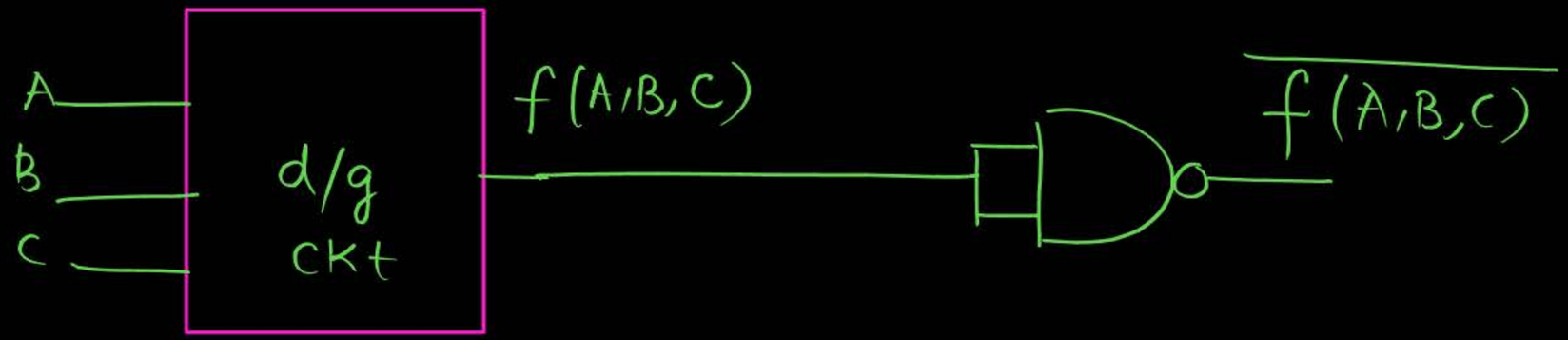
- o/p will be '0' only if all the i/p lines are '1' together.

- 

NAND As an Universal Gate

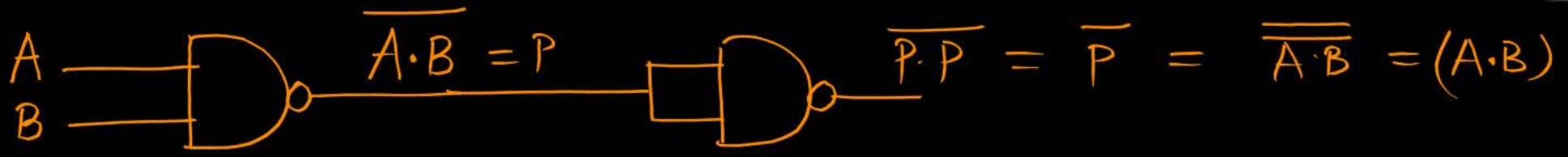
- NOT GATE :

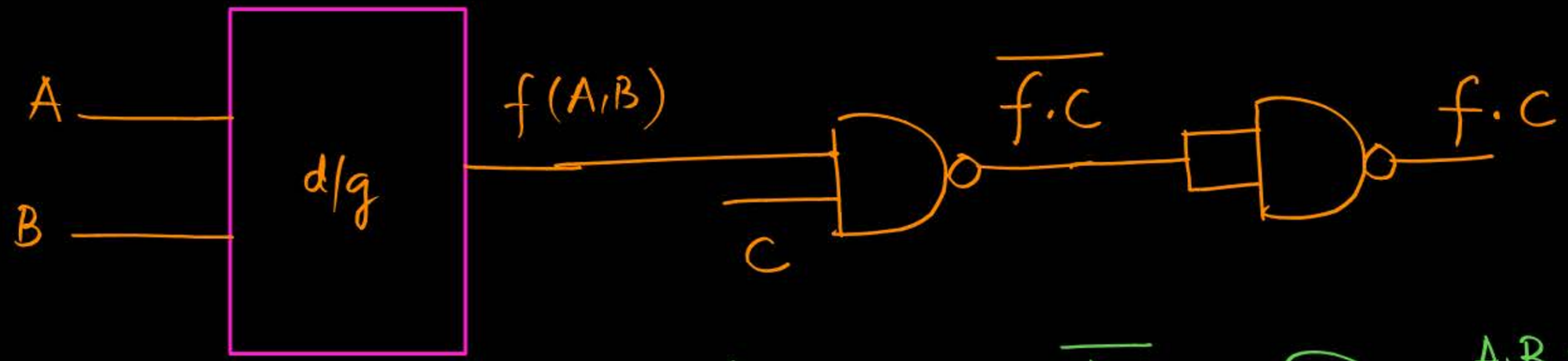




- **AND GATE :** 2 (2 i/p NAND) for 2 i/p AND operation.

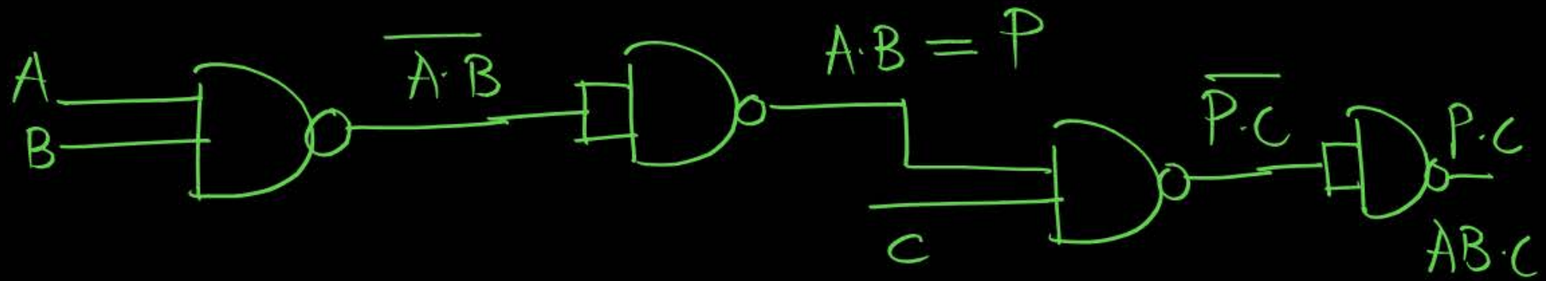
A, B $\longrightarrow$ A·B



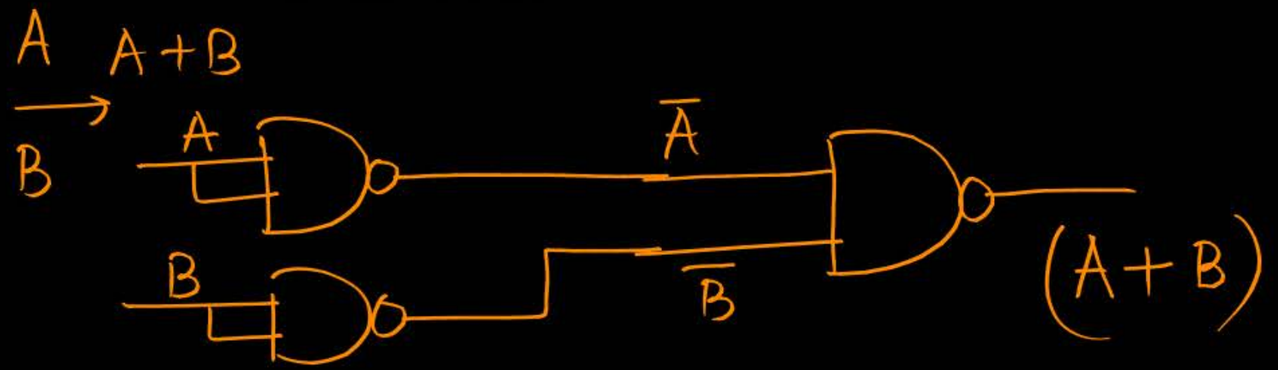


$\boxed{A \cdot B \cdot C}$

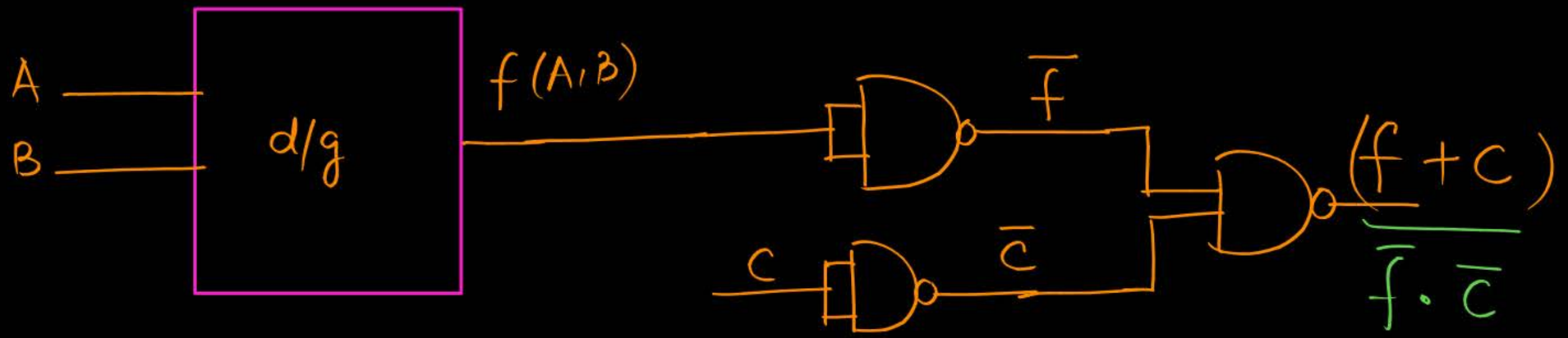
$P \cdot C$



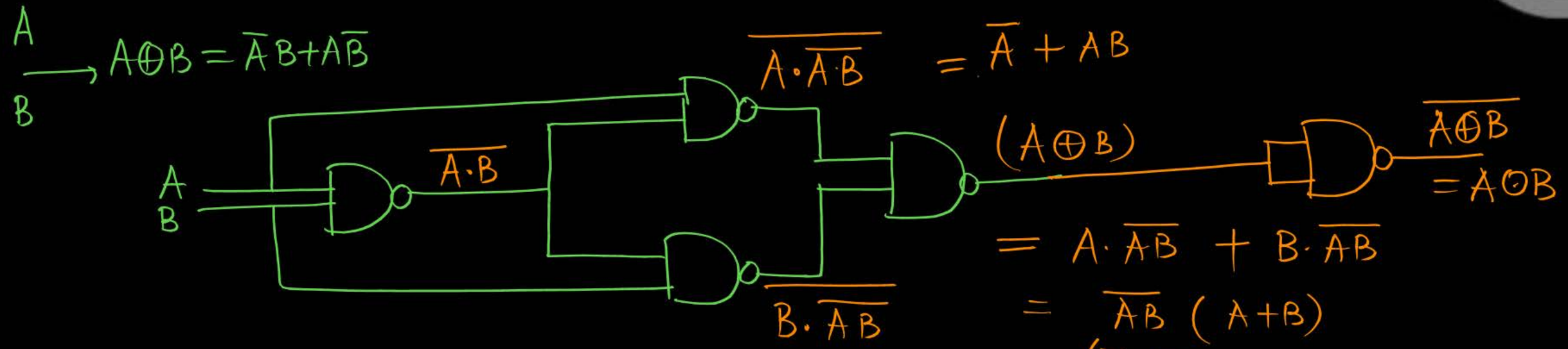
- **OR GATE :**



3 (2i/p NAND) for 2i/p OR operation

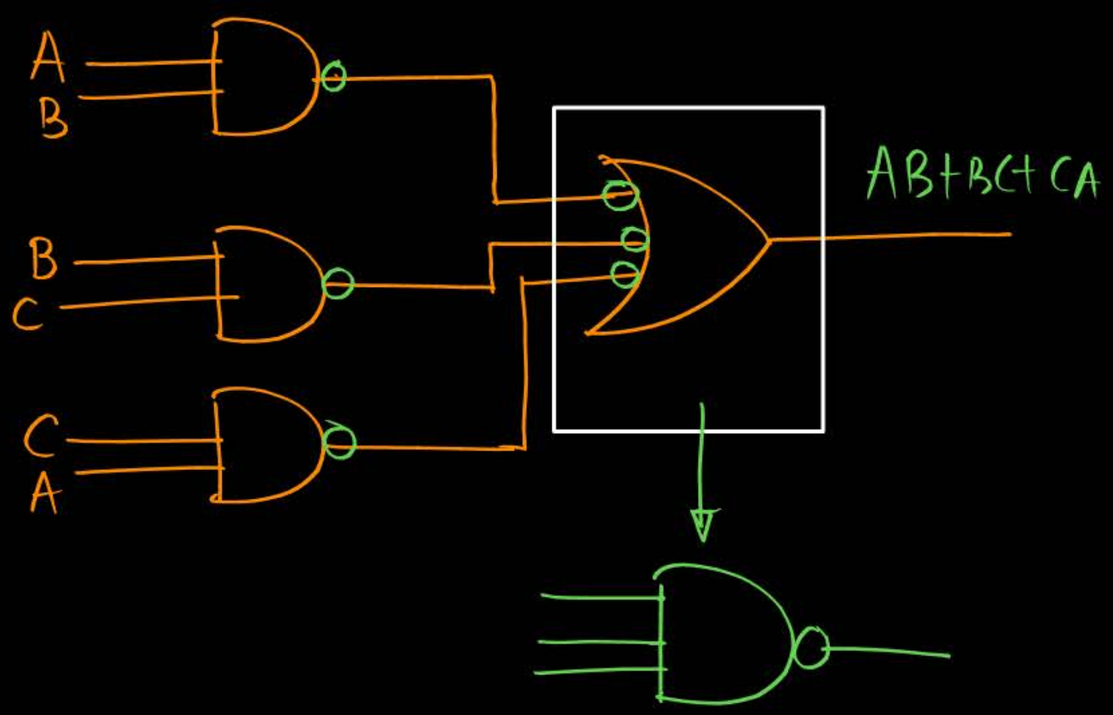


- **XOR GATE and XNOR GATE :**

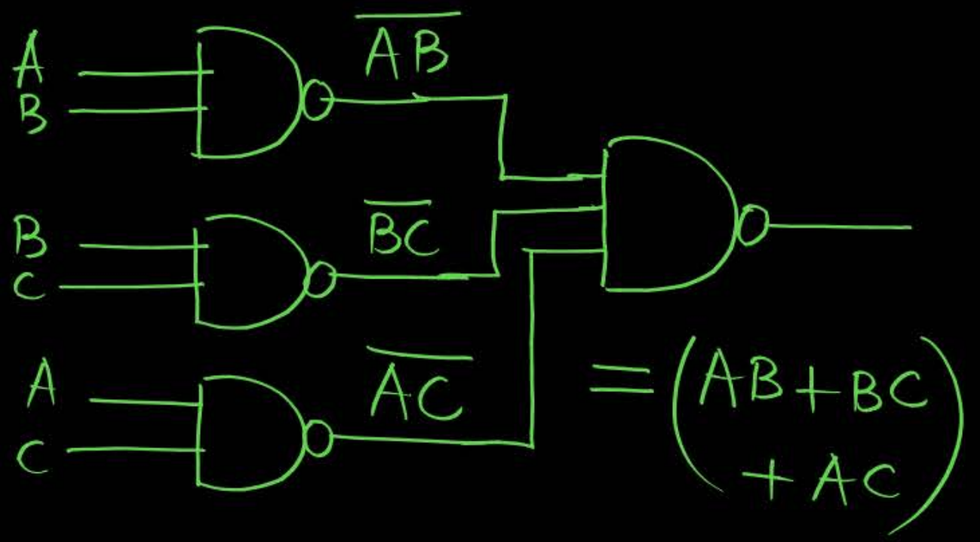


2-Stage Implementation of SOP using NAND Gate

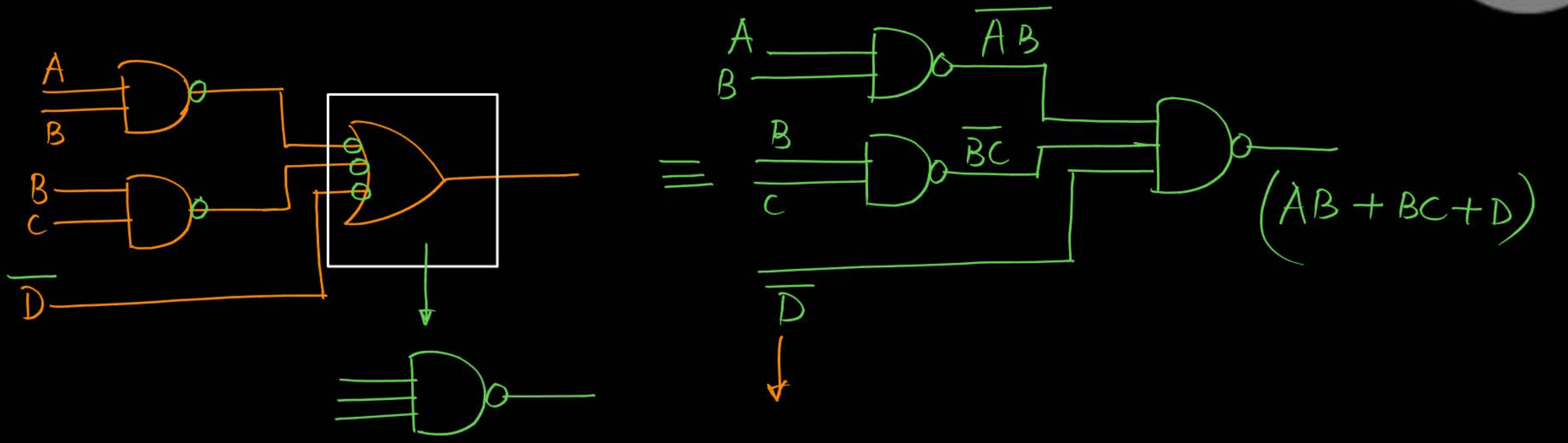
- $y = AB + BC + CA$



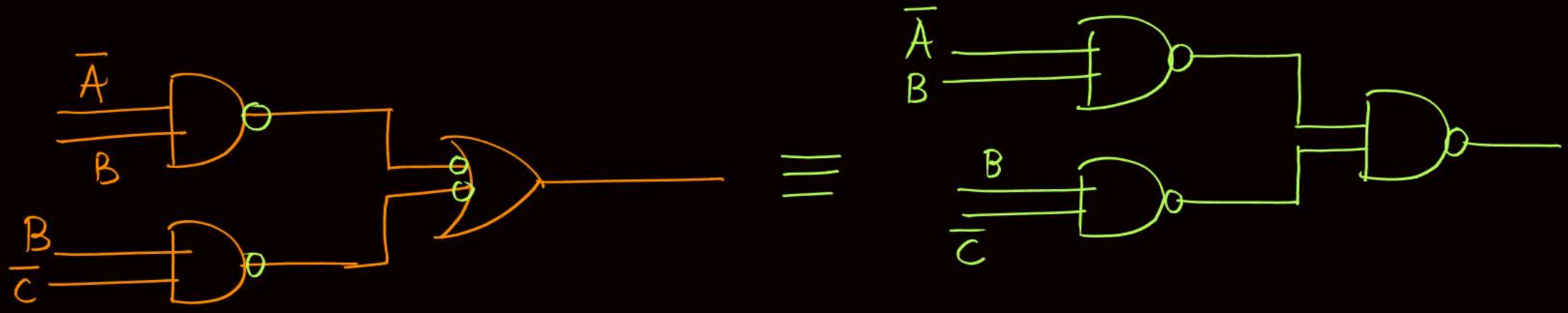
≡



- $y = AB + BC + D$



- $f(A,B,C) = \bar{A}B + B\bar{C}$



Question on minimum no. of NAND Gates

Question

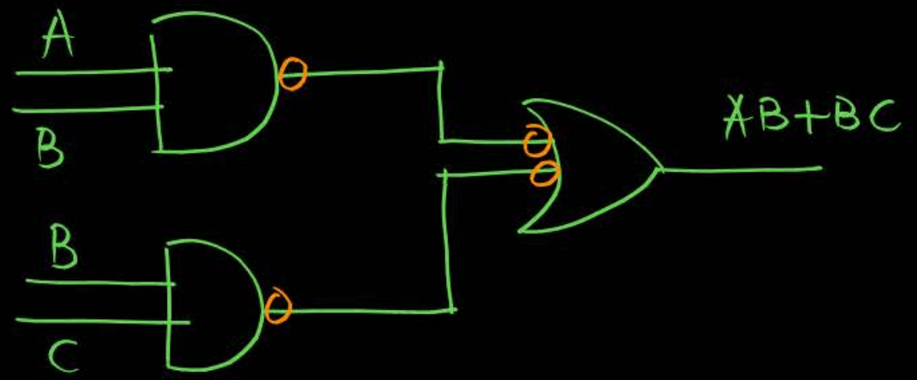
$y = AB + BC$, to implement y, minimum no. of 2-input NAND gate required is 3.

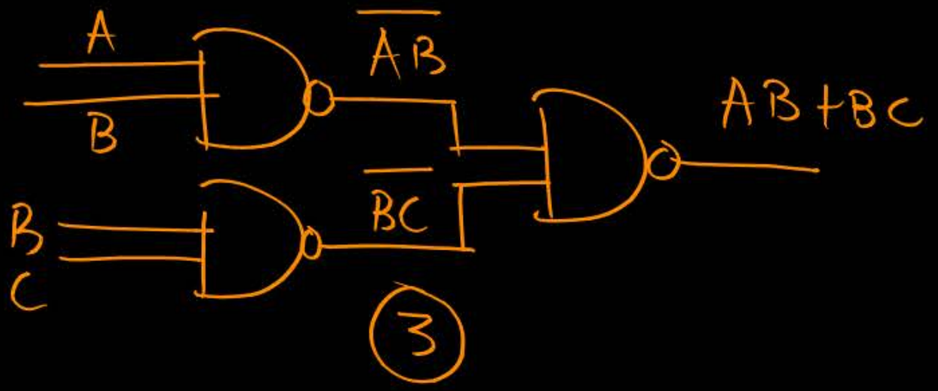
$Y = \underline{AB} + \underline{BC} = (P + Q)$ → 3 NAND

$A \cdot B \longrightarrow$ 2 NAND $\longrightarrow P$

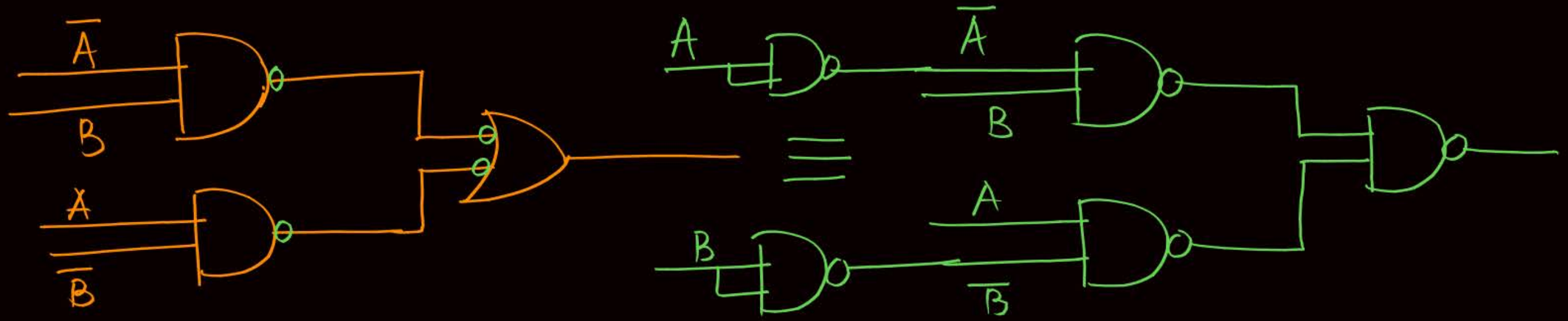
$B \cdot C \longrightarrow$ 2 NAND $\longrightarrow Q$

Total 7 NAND gates.





$$f(A,B) = \bar{A}B + A\bar{B}$$



$f(A,B,C) = (AB+BC)$

$\overline{f} = \overline{AB+BC} = \overline{AB} \cdot \overline{BC}$

$\overline{AB}$ → P (1 NAND)

$\overline{BC}$ → Q (1 NAND)

$\overline{f} = P \cdot Q$

$f = \overline{P \cdot Q}$ → (1 NAND)

OR → 3 NAND
$A \cdot B$ → 2 NAND
$\overline{A \cdot B}$ → 1 NAND

3

$f(A,B,C) = \bar{A}B + B\bar{C}$

$f(A,B,C) = B[\bar{A}+\bar{C}] = B \cdot \overline{AC}$

$= B \cdot P$ (2 NAND)

$\overline{AC}$ (P) → 1 NAND

$\bar{f} = \overline{\bar{A}\cdot B} \cdot \overline{B\cdot\bar{C}}$

$\overline{\bar{A}\cdot B}$ → 2 NAND → Q

$\overline{B\cdot\bar{C}}$ → (2 NAND) → R

$\bar{f} = Q \cdot R$

$f = \overline{QR}$ → 1 NAND

$f(A,B,C) = \bar{A}B + A\bar{B}$

$= (\bar{A}+\bar{B})(A+B)$

$= \overline{A\cdot B}\,(A+B)$

$f = (R)\cdot(A+B) = R\cdot A + RB$

(R): 1 NAND

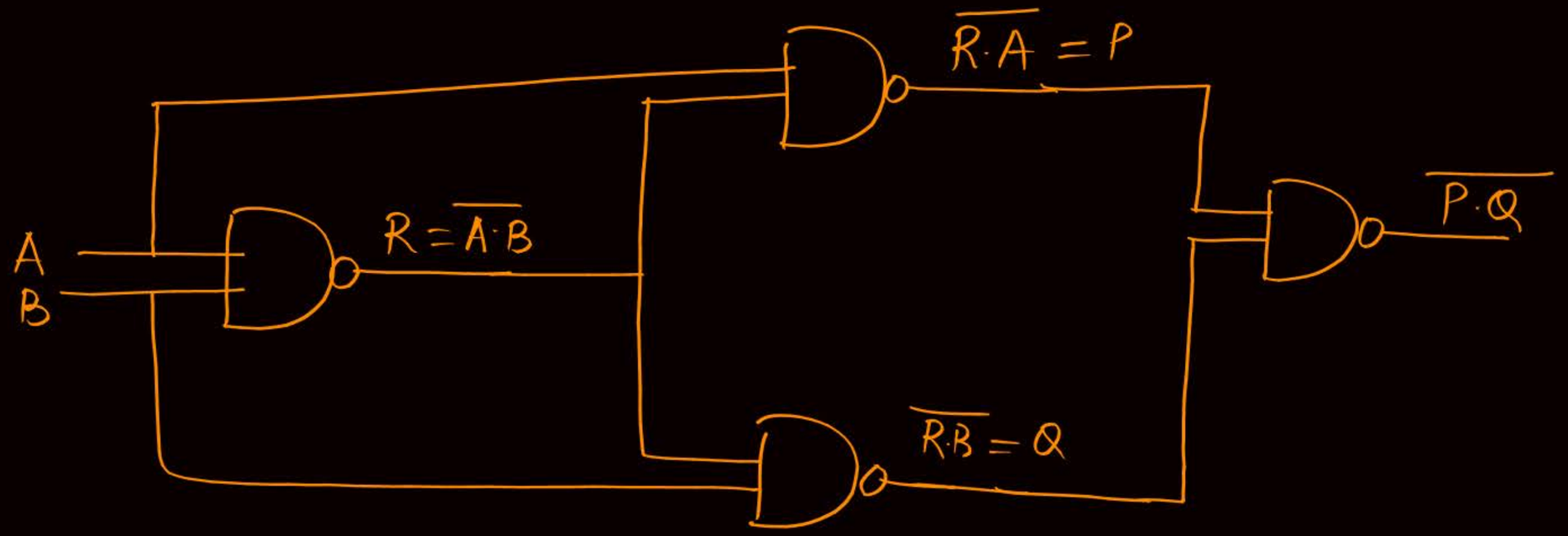
$\bar{f} = \overline{RA}\ \overline{RB}$

$\overline{RA}$ → P (1 NAND)

$\overline{RB}$ → 1 NAND (Q)

$\bar{f} = P \cdot Q$

$f = \overline{P\cdot Q}$ → 1 NAND


A
B
$R = \overline{A \cdot B}$
$\overline{R \cdot A} = P$
$\overline{R \cdot B} = Q$
$\overline{P \cdot Q}$

H·W

Minimum no. of 2 i/p NAND gate required to implement

1. $f(A,B,C) = A\bar{B} + B\bar{C} + A\bar{C}$

2. $f(A,B,C) = \bar{A}\bar{B} + \bar{B}\bar{C}$

3. $f(A,B,C) = AB + BC + AC$

4. $f(A,B,C) = \overline{AB\bar{C} \cdot D}$



2 Minute Summary

NAND gate:

Thank you

GW

Soldiers !

